

Job Posting: 178658 - Position: S26 Software Engineer (Embedded) 178658

Co-op Work Term Posted: 2026 - Summer
App Deadline 03/04/2026 09:00 AM
Application Method: Through UBC Science Co-op
Posting Goes Live: 02/25/2026 04:39 PM
Job Posting Status: Approved

ORGANIZATION INFORMATION

Organization Lumotive
Address Line 1 4664 Lougheed Hwy #162
City Burnaby
Postal Code / Zip Code V5C 3Y2
Province / State BC
Country Canada

JOB POSTING INFORMATION

Placement Term 2026 - Summer
** Job Title ** S26 Software Engineer (Embedded) 178658
Position Type Co-op Position
Job Location Burnaby, BC
Country Canada
Duration 8 months
Salary Currency CAD
Salary 3500.0 per month for 0 Major List

Job Description

Lumotive's revolutionary programmable optical semiconductors enhance perception, boost computing power, and ensure robust high-speed communication across various markets.

The award-winning, highly patented Light Control Metasurface (LCM™) chip is the first commercially available software-defined photonic beamforming solid-state technology addressing critical gaps in multiple industries, from 3D sensing to AI computing. Our LCM™ chip has already secured multiple design wins in our initial market, positioning it at the forefront of the 3D sensing evolution. By merging lidar and ToF camera technologies, it caters to a wide range of markets, including consumer electronics, robotics, industrial, and automotive sectors.

Lumotive was named Fast Company's Next Big Thing in Tech and won two CES Innovation Awards. Headquartered in Redmond, WA, with offices in San Jose, CA, and Vancouver, Canada, Lumotive is backed by notable investors, including Gates Frontier, MetaVC Partners, Quan Funds, Samsung Ventures, and Uniquist.

We are seeking a highly motivated Embedded Software Engineer (E1) to join our software development team. In this role, you will work closely with our technical teams to develop, optimize, and test embedded software for our next-generation LiDAR systems. This position offers an exciting opportunity to work on embedded systems, real-time processing, and higher-level applications such as perception, visualization, and user interfaces while contributing to industry-leading technology.

Responsibilities

- Develop and optimize embedded software and firmware for LiDAR sensors.
- Write, test, and debug C/C++ code for embedded processors.
- Assist in the development of real-time data acquisition and processing algorithms.
- Implement communication protocols (SPI, I2C, UART, CAN, Ethernet) for sensor interfacing.

- Collaborate with hardware engineers to integrate software with LiDAR hardware components.
- Participate in performance analysis, troubleshooting, and debugging of embedded systems.
- Develop software components for LiDAR perception, including object detection, tracking, and point cloud processing.
- Contribute to graphical user interface (GUI) applications for LiDAR data visualization and user interaction.
- Work on middleware and API development to facilitate integration with higher-level applications.
- Support automation scripts for LiDAR software builds and QA.

Job Requirements

Qualifications

Must have:

- Currently pursuing a Bachelor's or Master's degree in Computer Engineering, Electrical Engineering, Computer Science, or related field. 0-2 years of experience in embedded software development.
- Proficiency in C and C++, with experience in Python for scripting and automation
- Understanding of RTOS concepts, microcontroller programming, and hardware-software interfaces.
- Knowledge of communication protocols such as SPI, I2C, UART, CAN, and Ethernet.
- Basic understanding of digital signal processing and real-time data acquisition.
- Familiarity with Source code version control (Git).
- Ability to work collaboratively in a fast-paced, team-oriented environment.
- Strong problem-solving skills and attention to detail.

Preferred:

- Experience with FPGA development
- Exposure to LiDAR, computer vision, or 3D sensing technologies.
- Familiarity with Python for scripting and automation.
- Hands-on experience with hardware debugging tools (oscilloscopes, logic analyzers).
- Knowledge of Linux-based embedded systems development.
- Familiarity with Atlassian tools (Jira, Confluence, Bitbucket).
- Familiarity with working in Agile environments (Scrum / Kanban)

Citizenship Requirement N/A

APPLICATION INFORMATION

Application Procedure	Through UBC Science Co-op
Cover Letter Required?	Yes
Address Cover Letter to	Raymond To, Recruiter